

LOW-COST ATTITUDE DETERMINATION FOR CUBESATS USING SENSOR FUSION

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ABSTRACT

This paper presents the implementation and characterization of a Multiplicative Extended Kalman Filter (MEKF) for CubeSat attitude determination, integrating photodiode-based sun sensors, magnetometer, and inertial measurement unit (IMU) data. Building on previous work that established a two-vector TRIAD algorithm using sun and magnetic field vectors, this semester focused on achieving continuous attitude estimation through sensor fusion. The MEKF was selected as the optimal filtering approach due to its proven performance in spacecraft applications and its ability to handle sensors with different noise characteristics and update rates. Simulations validated the filter's performance with a cubesat spinning at 5 deg/s, processing IMU data at 100 Hz and TRIAD measurements at 1 Hz. Significant hardware improvements were made to address photodiode noise issues through circuit modifications including ground plane corrections, power supply filtering with RC low-pass filters, and input capacitance additions to eliminate ADC crosstalk. Comprehensive characterization testing was conducted to quantify sensor performance, revealing magnetometer standard deviations of approximately 0.0018 rad/s and highly stable photodiode readings after circuit corrections. Static TRIAD testing demonstrated angle measurement accuracy within 3 degrees for most orientations, with challenges remaining for certain angles likely due to calibration issues. The system was transitioned to a wireless configuration with integrated battery power and redesigned mechanical structure in preparation for dynamic testing. A multi-threaded software architecture was developed to enable real-time MEKF operation with live sensor data. This work establishes the foundation for implementing complete attitude determination and control capabilities in future iterations.

NOMENCLATURE

ADCS	Attitude Determination and Control System
ADC	Analog-to-Digital Converter
EKF	Extended Kalman Filter
MEKF	Multiplicative Extended Kalman Filter
UKF	Unscented Kalman Filter
IMU	Inertial Measurement Unit
TRIAD	TRIaxial Attitude Determination
I2C	Inter-Integrated Circuit
SPI	Serial Peripheral Interface
ZRO	Zero Rate Output
RMS	Root Mean Square
MPU 6050	Six-axis IMU (Gyro + Accelerometer)
AK09940	Three-axis Magnetometer
MCP3008	10-bit Analog-to-Digital Converter

1. INTRODUCTION

Attitude determination is a critical component of any spacecraft, allowing the satellite to know its orientation in space for communications, payload operations, and power generation. For CubeSats with limited size, weight, power, and budget, developing reliable attitude determination systems is challenging. Using multiple complementary sensors together through sensor fusion provides a practical way to achieve good attitude knowledge while keeping the system simple and affordable.

This work builds upon the photodiode-based sun sensor system developed last semester [1], which demonstrated two-vector attitude determination using the TRIAD algorithm. That system could calculate attitude when both sun and magnetic field vectors were available, but it couldn't provide continuous attitude knowledge between measurements or properly account for sensor uncertainties. The main goal this semester was to implement a Multiplicative Extended Kalman Filter (MEKF) to combine data from three sensors: photodiode sun sensors, a three-axis magnetometer (AK09940), and a six-axis IMU (MPU6050). This sensor fusion approach provides continuous attitude estimation at 100 Hz while accounting for measurement uncertainties.

The MEKF is the standard approach for spacecraft attitude estimation and has been used on missions including the Hubble Space Telescope and Mars Reconnaissance Orbiter [2]. Its main advantage is handling the multiplicative error representation for quaternion-based attitude, avoiding singularities found in other representations while being computationally efficient for systems with limited processing power. The filter combines high-rate gyroscope measurements with lower-rate absolute attitude references from sun and magnetic field observations, providing continuous attitude propagation without unbounded drift.

This semester's work covered several areas. First, research into sensor fusion techniques led to selecting and implementing the MEKF algorithm, including developing process and measurement noise models based on sensor specifications. Second, simulations validated the filter's performance with the cubesat rotating at 5 deg/s. Third, significant effort went into characterizing and improving the hardware, particularly fixing photodiode circuit noise issues from last semester. Fourth, testing protocols were developed to measure sensor performance and validate TRIAD accuracy through static rotation tests. Fifth, the system was redesigned for wireless operation with battery integration and a laser-cut acrylic cube structure. Finally, the software was redesigned to support real-time operation with live sensor data using multiple threads.

2. BACKGROUND

2.1 State Estimation and Kalman Filtering

State estimation involves determining a system's internal variables based on noisy or incomplete observations. The Kalman filter, introduced by Rudolf Kalman in 1960, provides an optimal solution for linear systems by combining predictions from a system model with measurements, weighting each based on their uncertainties [3]. The filter has two phases: prediction, where the state is propagated forward using the system model, and update, where the prediction is corrected using new measurements. This process repeats, providing optimal state estimates for linear systems with Gaussian noise.

For nonlinear systems like spacecraft dynamics, several extensions have been developed. The Extended Kalman Filter (EKF) linearizes the nonlinear models using first-order Taylor series expansions. This works well for many problems but can struggle when the nonlinearities are large [4]. The Unscented Kalman Filter (UKF) uses a sampling technique with sigma points to better capture nonlinear behavior, often giving better results at higher computational cost [5].

The three main Kalman filter types differ in how they handle models, noise, prediction, and updates. The Linear Kalman Filter works for linear systems. The Extended Kalman Filter extends this to nonlinear systems through linearization. The Multiplicative Extended Kalman Filter is a specialized version optimized for attitude estimation that handles the constraints of rotation representations.

2.2 MEKF for Attitude Estimation

Attitude representation has unique challenges because of constraints in rotation groups. Quaternions provide a four-parameter representation of attitude without singularities, but the unit norm constraint ($q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$) makes them difficult to use with standard filtering techniques. The Multiplicative Extended Kalman Filter (MEKF) solves this by representing attitude errors multiplicatively rather than additively [2]. Instead of estimating the full attitude quaternion, the MEKF estimates a small three-parameter error rotation vector and uses this to correct a separately propagated reference quaternion. This maintains the quaternion constraint naturally and results in a smaller, more efficient state vector.

The MEKF is the standard for spacecraft attitude estimation for several reasons. First, it maintains the quaternion constraint without needing normalization corrections that can introduce errors. Second, the three-parameter error representation avoids singularities in Euler angles while eliminating quaternion redundancy. Third, the small angle assumption for the error is typically valid when measurements arrive frequently, leading to accurate linearizations. Fourth, computational requirements are modest compared to alternatives like the UKF, making it suitable for limited spacecraft processors [6].

The filter works by propagating the attitude quaternion forward using gyroscope measurements, accumulating errors from biases and noise. When absolute attitude measurements become available from sensors like sun sensors or magnetometers (processed through TRIAD), these correct the attitude estimate based on the relative uncertainties. The gyroscope biases are estimated simultaneously, allowing the filter to adapt to slow drifts from temperature changes or aging. This is critical for

long-term accuracy since uncorrected bias causes unbounded attitude errors.

The MEKF state vector has six elements: three for the attitude error rotation vector ($\delta\theta$) and three for the gyroscope bias estimate (b). The attitude error is the small rotation from estimated to true attitude. The error dynamics are driven by gyroscope noise and bias drift, while the bias is modeled as a random walk to capture temperature variations and other slow changes.

2.3 Sensor Characteristics and Fusion Strategy

The three sensors provide complementary information with different characteristics. The photodiode sun sensors give absolute attitude information when the sun is visible, with measurements at about 1 Hz due to computational and communication limits. As described in previous work [1], these sensors have minimal drift but can be affected by ambient light and have limited accuracy, typically a few degrees. The six photodiodes (one per cube face) detect light intensity that varies with the cosine of the angle between incident light and the photodiode normal, allowing calculation of a sun vector in the body frame.

The three-axis magnetometer (AK09940) similarly provides absolute attitude information at 1 Hz, measuring Earth's magnetic field vector with standard deviations around 0.002 rad/s based on testing. Both sensors require knowledge of reference vectors in the inertial frame, computed from orbital position using standard models. The World Magnetic Model (WMM) provides magnetic field direction at any location, while solar position comes from time and orbital parameters. These two vectors are combined using TRIAD to produce attitude quaternion measurements.

The six-axis IMU (MPU6050) provides gyroscope and accelerometer measurements at 100 Hz. For attitude determination, the gyroscope is primary, providing angular rate information for continuous attitude propagation between 1 Hz absolute measurements. According to specifications, the gyroscope has a Zero Rate Output (ZRO) of ± 0.5 deg/s and bias drift of 0.015 deg/s per degC. For a small satellite with thermal variations of ± 0.2 deg/s from orbital heating and cooling, this means bias drift rates requiring continuous estimation. The total RMS noise at 100 Hz is about 0.038 deg/s from the datasheet, though actual testing showed different values used in filter tuning.

The sensor fusion strategy uses these complementary characteristics. High-rate gyroscope data provides smooth, continuous attitude propagation with low noise but accumulated drift from bias errors. Lower-rate absolute measurements provide drift-free references but with higher noise and possible unavailability (sun occultation, magnetic disturbances). The MEKF optimally combines these, using absolute measurements to bound gyroscope drift while using gyroscope data to smooth noisy absolute measurements and provide 100 Hz attitude knowledge. Each sensor's weaknesses are compensated by the others' strengths.

2.4 Triad Algorithm Integration

The TRIAD algorithm bridges raw sensor measurements and the MEKF, converting sun and magnetic field observations into attitude quaternions for the filter [7]. As described in previous work [1], TRIAD constructs two orthonormal triads from measured vectors in both body frame (from sensors) and reference frame (from models). The algorithm normalizes the two input vectors, constructs the first triad vector from the first input (typically sun), the second from the cross product of the two inputs, and the third from the cross product of the first two. This is done in both frames, and the rotation matrix between frames is computed. The rotation matrix is then converted to a quaternion for the MEKF.

This algorithm is computationally efficient and guaranteed to produce a valid rotation matrix, but doesn't optimally account for measurement uncertainties like more sophisticated algorithms (QUEST or Davenport's q-method) [8]. However, TRIAD works well for providing measurements to a Kalman filter that can properly weight observations. TRIAD implicitly weights the first vector more heavily, which is appropriate when that vector (sun) is more accurate than the second (magnetic field).

The integration creates a complementary architecture where TRIAD processes 1 Hz sun and magnetic field measurements to produce attitude quaternions as MEKF measurement inputs. The filter compares these TRIAD-derived attitudes with propagated gyroscope attitude, computing an innovation that drives the filter update. The measurement noise covariance for TRIAD observations must account for uncertainties in both sensors, propagated through the algorithm. Initial simulations used a simplified 3-degree standard deviation model, with refinement based on actual characterization planned for future work.

3. DESIGN AND IMPLEMENTATION

3.1 MEKF Mathematical Framework

The MEKF implementation follows standard spacecraft attitude estimation formulation with adaptations for the specific sensors and platform. The state vector has six elements: $x = [\delta\theta^T, b^T]^T$, where $\delta\theta$ is the three-component error rotation vector and b is the three-component gyroscope bias estimate. The attitude error is the rotation from estimated to true attitude. The gyroscope bias captures the slowly varying offset that would otherwise cause unbounded drift.

The state propagation model uses gyroscope measurements to update the attitude quaternion through quaternion kinematics. The reference quaternion q is propagated using corrected angular rate $\omega_{corrected} = \omega_{measured} - b_{est}$, where $\omega_{measured}$ comes from the IMU and b_{est} is the current bias estimate. The quaternion propagation follows $dq/dt = 0.5 * \text{big_}\omega(\omega) * q$, where $\text{big_}\omega$ is the skew-symmetric matrix from the angular rate. For discrete-time implementation at 100 Hz ($dt = 0.01$ s), this is integrated using the exact solution forming a delta quaternion from angular displacement over the timestep.

The discrete-time prediction equations compute the prior state estimate and covariance at each timestep. The error state is

predicted to stay zero (as the reference quaternion absorbs propagation), while bias is modeled as a random walk: $b_k = b_{k-1} + w_b$, where w_b is process noise representing temperature-induced drift. The state transition matrix F captures linearized dynamics:

$$F = \begin{bmatrix} I - [\omega_{cross}] \delta t, & -\delta t I \\ 0, & I \end{bmatrix}$$

where $[\omega_{cross}]$ is the skew-symmetric cross-product matrix of corrected angular rate and I is the 3×3 identity matrix. The covariance prediction is $P_k = F * P_{k-1} * F^T + Q$, where Q is the process noise covariance containing both gyroscope measurement noise and bias drift rate.

The measurement model relates the state to TRIAD-derived attitude observations. When new sun and magnetic field measurements arrive, TRIAD computes an attitude quaternion q_{meas} as the measurement. The measurement equation relates true attitude to this observation through the error quaternion: $q_{true} = \delta q \otimes q_{est}$, where $\otimes$ is quaternion multiplication and δq is the quaternion from error vector $\delta\theta$. The innovation is computed as the quaternion difference between measurement and estimate, linearized to produce a three-component residual $z = 2 * \delta q_{vec}$, where δq_{vec} is the vector part of $\delta q = q_{meas} \otimes q_{est}^*$.

The measurement sensitivity matrix H is simply $H = [I, 0]$, showing the measurement directly observes attitude error but not bias. The Kalman gain is $K = P * H^T * (H * P * H^T + R)^{-1}$, where R is measurement noise covariance. The state update applies the gain to innovation: $\delta x = K * z$, providing corrections $\delta\theta$ and δb . The attitude correction is applied by forming a quaternion from $\delta\theta$ and multiplying with the reference quaternion. The bias correction is added directly. The covariance update follows the standard Joseph form for stability: $P = (I - K * H) * P$.

3.2 Simulation Development and Validation

To validate the MEKF before hardware integration, a simulation environment was developed in Python. The simulation models a CubeSat spinning about its z-axis at 5 deg/s, representative of a detumbling scenario. This rotation speed tests the filter's tracking ability while staying within typical ranges for small satellites. The true attitude is propagated using perfect quaternion kinematics, providing ground truth for evaluation.

Simulated sensor measurements include realistic noise based on datasheet specifications. IMU data is generated at 100 Hz with true angular velocity corrupted by three sources. First, constant gyroscope bias initialized to $[0.3, -0.2, 0.25]$ deg/s representing the ZRO specification of ± 0.5 deg/s. Second, random walk bias drift with standard deviation of 0.015 deg/s per temperature change, modeling the 0.015 deg/s per degC specification with expected thermal variations of ± 0.2 degC/s. Third, white Gaussian noise with RMS value of 0.038 deg/s representing the sensor's intrinsic noise. The resulting measurement is $\omega_{measured} = \omega_{true} + b_{gyro} + b_{drift} + \nu_{gyro}$.

TRIAD measurements are simulated at 1 Hz, beginning 60 seconds into simulation to represent time for sun acquisition and initialization. The simulation initially runs on pure gyroscope integration to let bias errors accumulate, showing the need for absolute measurements. The noise model for TRIAD applies a random rotation of ± 3 degrees about a random axis to the true attitude quaternion, representing expected accuracy degradation from sensor errors through the algorithm. This multiplicative noise correctly captures attitude errors as rotations: $q_triad = q_truth \otimes q_noise$. Simulation results confirmed the MEKF performs as expected. The filter successfully estimates both attitude and gyroscope bias, with attitude errors converging within bounds defined by measurement noise. After TRIAD measurements begin at 60 seconds, the filter rapidly corrects accumulated drift and maintains accurate attitude knowledge with errors bounded by TRIAD noise. Bias estimates converge toward true values, showing the filter's ability to learn and compensate for sensor imperfections. A complementary filter was also implemented using a fixed weighting factor ($\alpha = 0.95$) to blend gyroscope and TRIAD data. However, the complementary filter shows significantly worse performance for attitude error and does not have a way of estimating bias. Data for both filters during the simulation can be seen in Figure 1 and Figure 2.

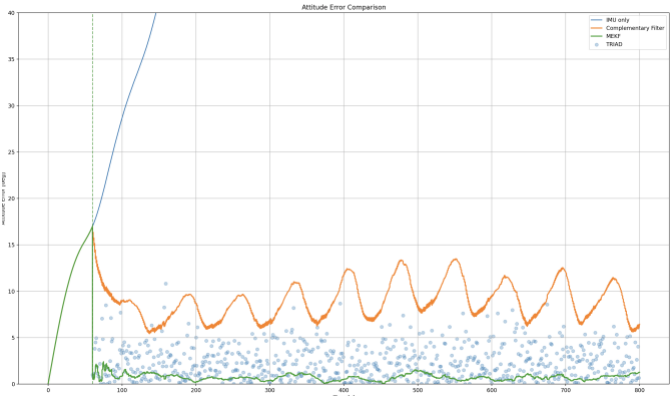


Figure 1: Attitude Error vs. Time

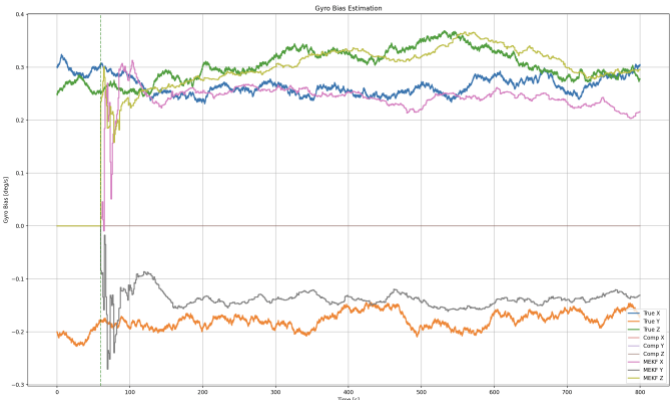


Figure 2: Estimated Bias vs. Time

3.3 Circuit Improvements

Significant effort focused on resolving photodiode noise issues from previous work [1]. Initial testing showed several noise patterns: synchronized jumps affecting all photodiodes, wave-like patterns propagating from channel to channel, and occasional voltage offsets. Systematic investigation using oscilloscope analysis and iterative modifications led to identifying and correcting three root causes. The first issue was a resistor between ground rails intended to separate analog and digital grounds. While meant to reduce Raspberry Pi digital noise affecting analog photodiode signals, the resistor caused voltage drops when current flowed, shifting the ADC ground reference above 0V. This threw off all photodiode readings, causing synchronized jumps. The solution was simple: removing this resistor and connecting ground rails directly for a common reference. The second issue was noise on the ADC's power supply rail (VDD). The Raspberry Pi's 3.3V output can have noise from USB communication, GPIO switching, and processor frequency changes. This couples into the ADC's reference voltage, causing measurement variations. An RC low-pass filter was added to the VDD line: an 18ohm resistor followed by two parallel capacitors. A 100muF electrolytic capacitor handles low-frequency noise and provides bulk storage, while a 0.1muF ceramic capacitor responds to high-frequency transients. The 100muF capacitor has higher capacitance but slower response due to higher ESR, while the 0.1muF ceramic has lower ESR for quick high-frequency response but less storage. Together they provide broadband filtering. A wiring diagram of this change can be seen in Figure 3.

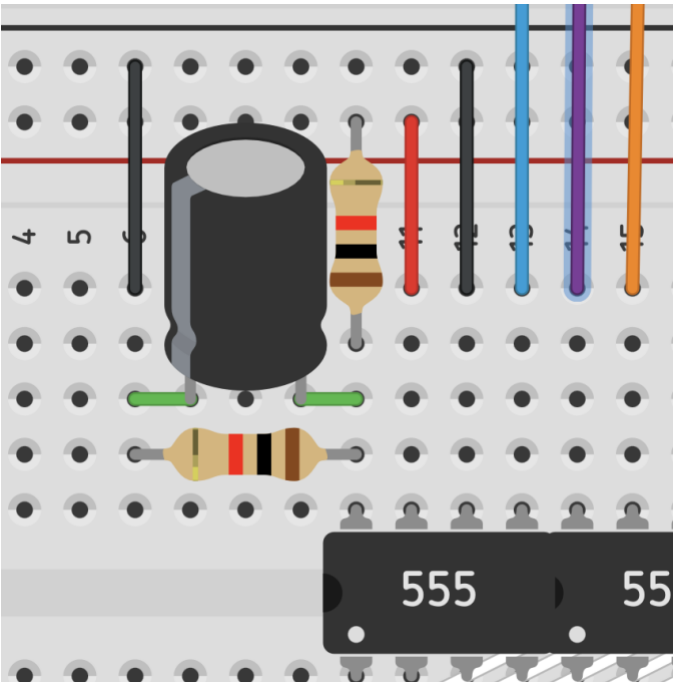


Figure 3: RC Low-pass Filter

The third issue was channel-to-channel crosstalk in the ADC. The MCP3008 uses an internal 20pF sample-and-hold capacitor that charges to the input voltage. When switching channels, this capacitor must discharge from the previous voltage and recharge through the high 10kohm photodiode impedance. The RC time constant of $20\text{pF} \times 10\text{kohm} = 200\text{ns}$ means if the ADC samples too quickly, the capacitor doesn't settle fully, causing previous channel voltage to bleed into the current measurement. This creates a wave pattern where each reading is contaminated by the previous channel.

The solution was adding 0.1μF capacitors at each ADC input, between the input pin and ground. These serve two purposes. First, they provide charge storage to rapidly charge/discharge the ADC's internal 20pF capacitor without pulling current through the high-impedance photodiode circuit, eliminating voltage drop during sampling. Second, they isolate each channel by providing a low-impedance path to ground at the ADC switching frequency, preventing crosstalk. The much larger 0.1μF capacitance compared to internal 20pF means the input capacitor voltage barely changes when supplying charge, ensuring stable measurements. A wiring diagram of the improved circuit can be seen in Figure 4.

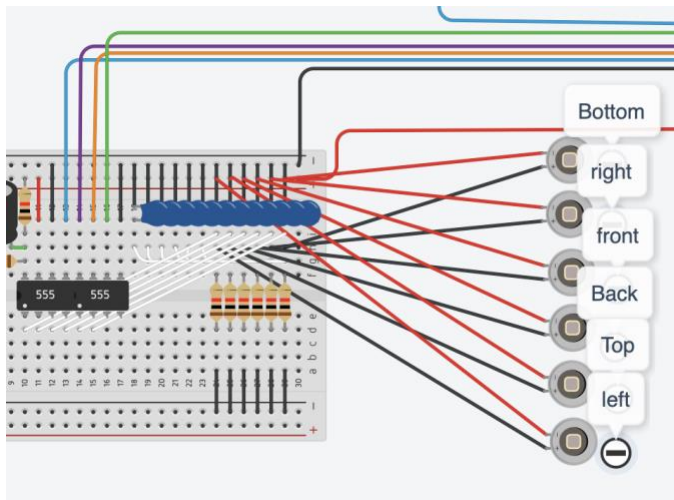


Figure 4: 0.1μF Capacitors Circuit Improvement

After implementing all three fixes, photodiode noise was dramatically reduced, as confirmed by testing. Readings became extremely stable with standard deviations approaching zero in many cases, showing the modifications successfully eliminated the main noise sources.

3.4 Sensor Characterization Testing

Comprehensive testing was conducted to measure magnetometer and photodiode performance, providing data for filter tuning. Professor Rosen suggested starting simple: rotating the cube manually on paper with a compass. This proved difficult ensuring rotation about the cube's center, and the rectangular box made certain angles inaccurate. A better solution was a drafting compass tool, with the top velcroed to the cube and bottom to the table, allowing controlled rotation

about the cube's center. The physical test setup is shown in Figure 5.

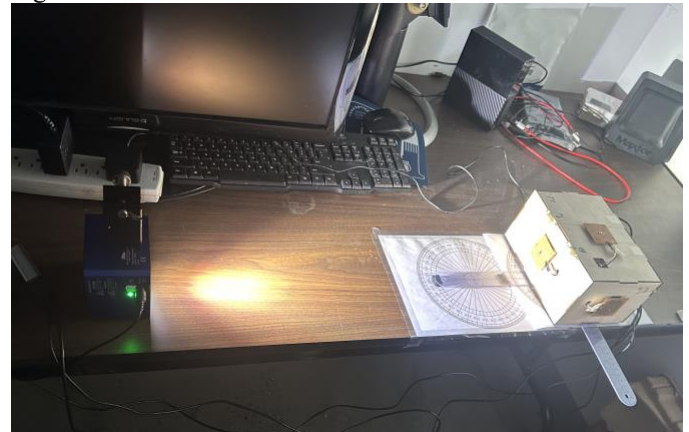


Figure 5: Static Test Setup

For the magnetometer, static testing collected about 250 readings with the cube on the table. Results showed excellent stability, confirming low noise. For photodiodes, testing was done in a dark room using the Ossila solar simulator to eliminate ambient light. Static testing with about 280 readings showed remarkably stable results as well after circuit modifications.

Static TRIAD testing measured the cube's attitude at different rotation angles: 0 deg, -45 deg, -90 deg, +45 deg, +90 deg, and returning to 0 deg. The compass tool allowed precise angle setting while the solar simulator provided consistent light. Results showed varying performance by angle.

Magnetometer data stayed stable across all angles with standard deviations in the 15-50 range (raw ADC units), showing the issue was specifically with photodiodes at that orientation. Interestingly, rotating from -90 deg to +90 deg, the magnetic field vector didn't flip as expected, showing magnetometer recalibration is needed since the experimental setup changed.

3.5 Mechanical System Redesign

The system underwent significant mechanical redesign for wireless operation and better testing. A 3D-printed cube replaced the previous laser-cut wooden version, with improvements including photodiodes centered on each face for optimal optics, while maintaining Raspberry Pi port access for programming and debugging. One face includes a mounting point for a servo motor for future dynamic testing. The design allows individual faces to be removed for maintenance, and dimensions roughly match a standard 1U CubeSat (10cm cube). Battery integration had several challenges. An initial attempt used a 12V NiMH 2000mAh battery from the lab with a buck converter to step down to 5V for the Raspberry Pi. However, testing showed the system shut down after only 5 minutes, showing either insufficient capacity or a battery/converter problem. This was abandoned for a commercial lithium-ion battery pack rated for 5V, 3A output with 10,000mAh capacity. This provides ample runtime for extended testing (theoretically 10-20 hours depending on processor load) while eliminating

voltage conversion complexity. The CAD and the physical box can be seen in Figure 6 and Figure 7.

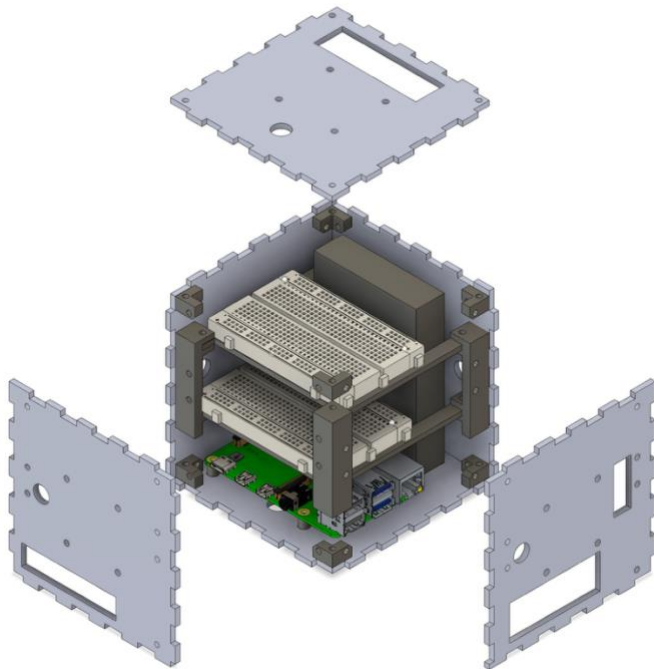


Figure 6: CAD of the Improved ADCS Box

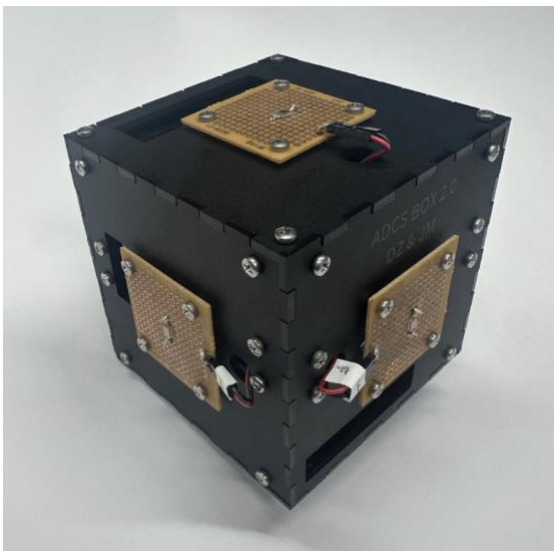


Figure 7: Physical Assembly of Improved ADCS Box

A stepper motor test stand was developed for dynamic characterization, mounting the cube to a stepper motor shaft for controlled rotation. The setup uses a smaller stepper motor than originally planned but should provide adequate torque for the lightweight cube. The motor is driven by a microstepper controller providing precise angular positioning and controlled rotation speeds. This enables several tests: measuring TRIAD accuracy as a function of rotation angle by stepping through

defined positions, characterizing dynamic performance by rotating at fixed angular velocities and comparing measured angles to known rotation, and quantifying measurement lag and offset errors. The stepper setup was assembled and briefly tested, confirming basic functionality, but full testing was deferred to next semester due to time constraints. The wiring diagram and physical setup for the stepper motor can be seen in Figure 8 and Figure 9.

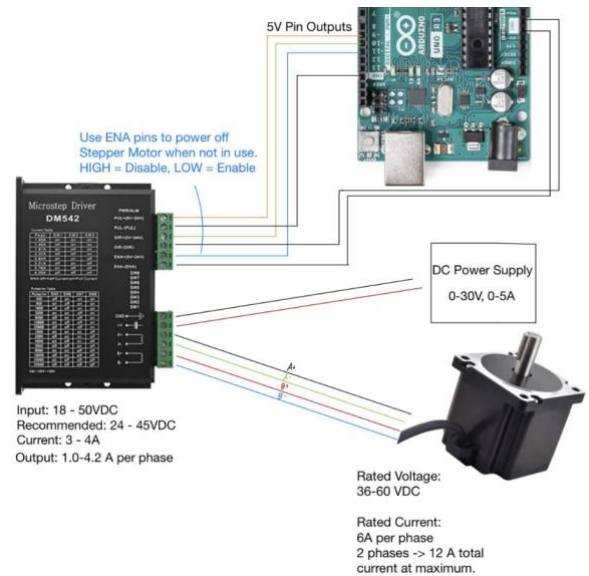


Figure 8: Stepper Motor Setup Diagram

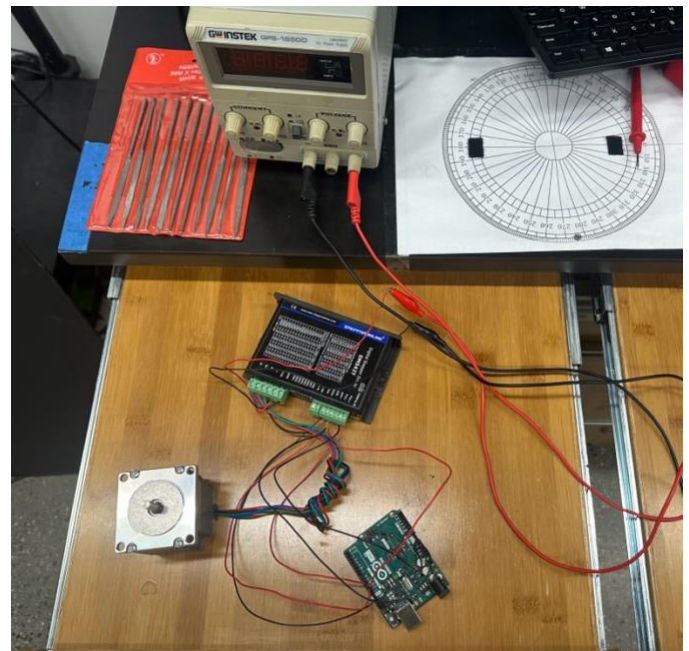


Figure 9: Physical Stepper Motor Setup

3.6 Real-Time Software Implementation

A critical challenge in transitioning from simulation to hardware was redesigning software for real-time operation with

live sensor data at different update rates. The original simulation code operated sequentially, processing one IMU timestep then checking for TRIAD updates, but this doesn't work when sensors operate asynchronously. The solution was a multi-threaded architecture separating data acquisition from processing.

Three concurrent threads were implemented using Python's threading module. The IMU thread runs continuously at 100 Hz, reading gyroscope data from the MPU6050 via I2C and storing it in a thread-safe data container protected by a mutex lock. The TRIAD thread runs at lower rate (about 10 Hz checking rate), calling the TRIAD algorithm which processes photodiode and magnetometer data when available and stores the resulting attitude quaternion in shared data. The filter thread runs at 100 Hz, implementing the main MEKF loop by reading latest sensor data, propagating the state, and performing measurement updates when new TRIAD data is flagged.

This architecture solves several problems. First, it ensures high-rate IMU data is never missed, as the IMU thread runs independently of filter processing time. Second, it allows TRIAD processing (which can take 50-100ms due to ADC communication overhead) to occur asynchronously without blocking the filter. Third, it provides clean separation between hardware interface code and filtering algorithms, improving modularity and testability. Thread-safe shared memory using locks ensures data isn't corrupted when multiple threads access it.

The filter thread implements the full MEKF algorithm from Section 3.1, propagating attitude using corrected gyroscope measurements at 100 Hz and performing measurement updates when TRIAD signals new data. For comparison, a complementary filter is also implemented using a fixed weighting factor ($\alpha = 0.95$) to blend gyroscope integration with TRIAD measurements. Both filters log attitude quaternion estimates to CSV files for post-processing. Status information is printed to console every second, showing current attitude quaternion, estimated gyroscope bias in degrees per second, and whether TRIAD measurements are being received.

4. TESTING AND RESULTS

4.1 Simulation Results

The simulation environment provided valuable validation before hardware integration. The simulated scenario with the cubesat spinning about its z-axis at 5 deg/s demonstrated the filter's ability to track rotation. During the first 60 seconds before TRIAD measurements become available, both MEKF and IMU-only integration diverge from truth due to accumulating gyroscope bias errors. The separation visually demonstrates drift rate, matching the injected bias of 0.3 deg/s in x, -0.2 deg/s in y, and 0.25 deg/s in z.

At 60 seconds, TRIAD measurements begin arriving at 1 Hz. The MEKF rapidly corrects accumulated error, with attitude estimates converging to within about 3 degrees of truth within a few measurement cycles. This convergence time is consistent with the filter gain and measurement noise levels. After convergence, the MEKF maintains bounded errors oscillating

around true attitude with amplitude determined by the TRIAD measurement noise of ± 3 degrees. IMU-only integration continues drifting unbounded, clearly showing the necessity of absolute measurements for long-term attitude knowledge.

Gyroscope bias estimation performance showed successful convergence. The MEKF successfully estimates three-axis bias, with all components converging toward true values after TRIAD measurements begin. Convergence is slower than attitude error convergence, as bias isn't directly observable and must be inferred from repeated attitude corrections. The x-axis bias converges from initial estimate of 0 to about 0.28 deg/s (truth: 0.30 deg/s), y-axis to -0.18 deg/s (truth: -0.20 deg/s), and z-axis to 0.23 deg/s (truth: 0.25 deg/s). The remaining small errors of 0.02-0.03 deg/s are consistent with observability limits imposed by 3-degree TRIAD noise.

4.2 Hardware Characterization Results

Magnetometer characterization provided baseline performance metrics for filter tuning. The sensor showed excellent stability when stationary, with standard deviations of [0.00178, 0.00117, 0.00169] in normalized units across about 250 samples. These correspond to about 0.1-0.2% of signal magnitude, showing high precision. The mean values of [-0.706, 0.105, 0.701] represent the normalized magnetic field vector at Cooper Union's location, which can be verified against IGRF model predictions. The consistency confirms the magnetometer operates correctly and has much lower noise than initially estimated, suggesting MEKF measurement noise covariance could be reduced from the initial 3-degree assumption to potentially 1-2 degrees for improved filter performance.

Photodiode characterization after circuit modifications showed dramatically improved performance compared to last semester [1]. Standard deviations were essentially zero ($[0.0, 1.11 \times 10^{-16}, 0.0]$), limited only by numerical precision in floating-point calculations. This represents complete elimination of noise issues from earlier testing, validating the circuit analysis and modifications. The mean sun vector of [-0.039, -0.999, 0.000] correctly shows the solar simulator was positioned approximately along the negative y-axis of the cube, with a small x-component suggesting slight misalignment or possibly a systematic calibration offset in one photodiode.

Static TRIAD testing revealed both strengths and challenges in two-vector attitude determination. Table 1 summarizes results at different rotation angles. Most angles showed good accuracy with errors of 2-3 degrees and very low standard deviations, consistent with expectations for photodiode-based sun sensors [1]. However, the -45 deg position showed significantly degraded performance with mean error of -41.081 deg and standard deviation of 2.916 deg. Investigation showed this was due to noise in sun vector measurements at that orientation, possibly from a loose connection later discovered and fixed.

Table 1: Static TRIAD Testing Results

True Angle (deg)	Measured Angle (deg)	Error (deg)	Std Dev (deg)
0	2.261	+2.261	0
-45	-41.081	+3.919	2.916

-90	-92.603	-2.603	0
+45	50.064	+5.064	0
+90	92.525	+2.525	0.009
0 (return)	2.261	+2.261	0.005

The consistent 2-3 degree offset at 0 deg (measured at both beginning and end of the test sequence) suggests a systematic calibration error rather than random noise. This could come from several sources: misalignment between compass rose markings and actual cube orientation, small errors in solar simulator position relative to assumed reference direction, or systematic errors in individual photodiode calibration. The excellent repeatability (returning to the same value after a full rotation sequence) shows the system is mechanically stable and measurements are deterministic, which is encouraging for filter performance.

The observation that the magnetic field vector doesn't flip when rotating ± 180 deg shows magnetometer recalibration is necessary. This is expected, as the experimental setup has been modified multiple times since original calibration. The magnetometer measures total magnetic field, including both Earth's field and local distortions from ferromagnetic materials or current-carrying wires. Changes to mechanical structure, battery and wiring addition, or even repositioning the test setup within the lab could all affect calibration. Future work should include systematic magnetometer calibration following established methods for hard-iron and soft-iron correction [9].

5. Conclusion and Future Work

This semester successfully established the foundation for continuous attitude determination in the CubeSat ADCS through implementing a Multiplicative Extended Kalman Filter integrating photodiode sun sensors, magnetometer, and IMU data. The MEKF was selected as the optimal approach after researching alternatives including standard EKF and UKF, with MEKF's multiplicative error representation providing ideal characteristics for spacecraft attitude estimation. Simulation studies validated the filter's ability to fuse high-rate gyroscope measurements with low-rate absolute attitude references, demonstrating convergence to 3-degree accuracy and successful gyroscope bias estimation.

Significant hardware improvements were achieved through systematic investigation and correction of photodiode circuit noise issues. Three separate root causes were identified and resolved: ground loop voltage drops from inappropriate resistor placement, power supply noise coupling through ADC reference voltage, and channel-to-channel crosstalk from insufficient input capacitance. The resulting system shows dramatically improved measurement stability, with photodiode standard deviations approaching numerical precision limits and magnetometer performance exceeding initial specifications. This measurement quality improvement directly enhances attitude determination accuracy and filter performance.

Comprehensive characterization testing quantified sensor performance under static conditions, providing data for proper filter tuning and validating hardware improvements. Static

TRIAD testing demonstrated 2-3 degree accuracy for most orientations, with systematic offsets showing calibration opportunities rather than fundamental limitations. The mechanical system was redesigned for wireless operation with integrated battery power and improved structure, enabling more flexible testing scenarios. A multi-threaded software architecture was developed to support real-time MEKF operation with live sensor data, separating data acquisition from filtering and allowing sensors operating at different rates to be properly fused.

5.1 Remaining Challenges

Several challenges were identified that need attention in future work. First, magnetometer calibration is needed to correct for hard-iron and soft-iron distortions from the modified mechanical structure and onboard electronics. The observation that the magnetic field vector doesn't reverse when rotating 180° clearly shows calibration drift. Standard two-step calibration procedures should be implemented, involving data collection while rotating the cube through all orientations followed by ellipsoid fitting to determine correction parameters [9].

Second, the systematic 2-3 degree offset in TRIAD measurements suggests photodiode calibration refinement. While current accuracy is acceptable for many CubeSat missions, improvement is possible through individual photodiode characterization to account for variations in sensitivity, spectral response, and mounting alignment [10].

Third, the live MEKF implementation needs further testing and debugging. While the threaded architecture is implemented, comprehensive testing with continuous operation wasn't completed due to time constraints and focus on characterization testing. Issues like cube visualization overhead still exist and need further development.

5.2 Future Work

The stepper motor test stand is ready for dynamic characterization testing to complement static results from this semester. Planned experiments include rotation at different fixed angular velocities while comparing TRIAD measurements to known rotation angle, enabling quantification of dynamic accuracy, measurement lag, and the effect of angular rate on attitude determination. This data will show whether TRIAD introduces systematic errors during motion and inform filter tuning for dynamic scenarios. The performance of the MEKF can also be dynamically tested using this setup. The upgraded 3D-printed cube can be mounted to the stepper motor for these tests, allowing controlled rotation while maintaining the wireless battery-powered operation demonstrated this semester.

The live MEKF implementation also needs further development, possibly separating the MEKF filtering from the cube visualization such that the visualization overhead doesn't affect the timing of the computations. The filter could output the raw quaternions to a csv file and a separate visualization code could be used to play back the motion.

Upgrading from TRIAD to the QUEST algorithm represents a natural next step for improved attitude determination. While

TRIAD is computationally efficient and works well for providing measurements to the MEKF, QUEST (QUaternion ESTimator) provides optimal weighting of the sun and magnetic field vectors based on their individual measurement uncertainties [8]. This would be particularly valuable given the characterization data showing different noise characteristics between the photodiodes and magnetometer. QUEST solves Wahba's problem to find the optimal attitude quaternion that minimizes a weighted sum of squared residuals, potentially improving accuracy by 1-2 degrees compared to TRIAD's hierarchical approach. The additional computational cost is modest and well within the Raspberry Pi's capabilities. Integration with a control system represents the next major milestone toward a complete ADCS. A magnetorquer-only attitude control system is currently under development and integrating it with the MEKF-based attitude estimation creates a closed-loop ADCS.

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